CHAPTER 6

PERMUTATIONS AND COMBINATIONS

CHAPTER ESSENCE AT A GLANCE:
Counting made effortless! Use Permutations (${}^{n}P_{r}$) when the ORDER of arrangement matters (like passwords, race positions, spelling letters). Use Combinations (${}^{n}C_{r}$) when order DOES NOT matter (like picking a cricket team, choosing committee members)!

4 CORE PRINCIPLES TO UNDERSTAND FIRST

- **Multiplication Principle:** If an operation can be performed in m ways and a second independent operation in n ways, sequential choices total m × n ways.
- **Permutations (Arrangements):** ${}^{n}P_{r}$ = n! over (n - r)! used when order matters (words, ranks, numbers). ${}^{n}P_{n}$ = n! and ${}^{n}P_{0}$ = 1.
- **Combinations (Selections):** ${}^{n}C_{r}$ = n! over [r! (n - r)!] used when order does NOT matter (committees, chords, teams). ${}^{n}C_{0}$ = 1 and ${}^{n}C_{n}$ = 1.
- **Key Identities:** ${}^{n}C_{r} = {}^{n}C_{n-r}$. If ${}^{n}C_{a} = {}^{n}C_{b}$, then either a = b or a + b = n. Pascal's Identity: ${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r}$.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$n! = n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1, \quad 0! = 1$$

Factorial Notation and Zero Factorial Convention

$${}^{n}P_{r} = \frac{n!}{(n-r)!} \quad (0 \le r \le n)$$

Permutation Formula: Ordered Arrangements of Distinct Objects

$${}^{n}C_{r} = \frac{n!}{r!\,(n-r)!} \rightarrow {}^{n}P_{r} = {}^{n}C_{r} \cdot r!$$

Combination Formula: Unordered Selection Multiplicity

$${}^{n}C_{r} + {}^{n}C_{r-1} = {}^{n+1}C_{r} \quad \text{(Pascal's Combinatorial Identity)}$$

Pascal's Recurrence Identity for Binomial Coefficients

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

- **Alert:** Remember 0! = 1 by definition, not 0! Also ${}^{n}C_{0}$ = 1 and ${}^{n}C_{n}$ = 1.
- **Alert:** Quick mental shortcut: ${}^{10}C_{2}$ is simply (10 × 9) over (2 × 1) = 45. Never compute full factorials by hand!

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. If nC8 = nC2, find the value of nC2. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Symmetric Property]** If nCa = nCb, then either a = b or a + b = n. Since 8 ≠ 2: n = 8 + 2 = 10. [1 Mark]

• **[Step 2: Calculate 10C2]** 10C2 = (10 × 9) over (2 × 1) = 90 over 2 = 45. [1 Mark]

• **[Boxed Result: n = 10, nC2 = 45]**

2. Find the number of 4-letter words, with or without meaning, that can be formed out of the letters of the word 'ROSE', where repetition of letters is not allowed. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Permutation Calculation]** The word 'ROSE' has 4 distinct letters (R, O, S, E). Number of arrangements of 4 letters taken all at a time = 4P4 = 4! = 4 × 3 × 2 × 1 = 24 words. [2 Marks]

• **[Boxed Result: 24 words]**

3. Derive the relation between Permutation and Combination: nPr = nCr x r!. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Combination Formula]** nCr = n! over [r! (n - r)!]. [1 Mark]

• **[Step 2: Multiplying by r!]** nCr x r! = [n! over (r! (n - r)!)] x r! = n! over (n - r)!. [1 Mark]

• **[Step 3: Recognizing nPr]** By definition of permutations: nPr = n! over (n - r)!. Therefore: nPr = nCr x r!. [1 Mark]

• **[Boxed Result: nPr = nCr x r! proved]**

4. Prove Pascal's Identity: nCr + nC(r - 1) = (n + 1)Cr. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Factorial Expansion]** LHS = n! over [r! (n - r)!] + n! over [(r - 1)! (n - r + 1)!]. [1 Mark]

• **[Step 2: Taking Common Factors]** LHS = n! over [(r - 1)! (n - r)!] [1 over r + 1 over (n - r + 1)] = n! over [(r - 1)! (n - r)!] [(n - r + 1 + r) over (r (n - r + 1))] = n! over [(r - 1)! (n - r)!] [(n + 1) over (r (n - r + 1))]. [1.5 Marks]

• **[Step 3: Recombining]** = (n + 1) n! over [r (r - 1)! x (n - r + 1)(n - r)!] = (n + 1)! over [r! (n + 1 - r)!] = (n + 1)Cr = RHS. Hence proved. [0.5 Mark]

• **[Boxed Result: nCr + nC(r-1) = (n+1)Cr (Pascal's Identity)]**

5. How many 3-digit even numbers can be made using the digits 1, 2, 3, 4, 6, 7 if no digit is repeated? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Units Place Selection]** For the number to be even, units place must be filled by an even digit from {2, 4, 6} (3 choices). [1 Mark]

• **[Step 2: Tens and Hundreds Places]** Total digits available = 6. After filling units place, 5 digits remain. Tens place can be filled in 5 ways. Hundreds place can be filled in 4 ways. [1 Mark]

• **[Step 3: Multiplication Principle]** Total 3-digit even numbers = 3 × 5 × 4 = 60 numbers. [1 Mark]

• **[Boxed Result: 60 even numbers]**

6. A committee of 7 has to be formed from 9 boys and 4 girls. In how many ways can this be done if the committee consists of: (a) Exactly 3 girls? (b) At least 3 girls? (c) At most 3 girls? [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: (a) Exactly 3 Girls]** Select 3 girls from 4 and (7 - 3) = 4 boys from 9: 4C3 × 9C4 = 4 x [(9 × 8 × 7 × 6) over (4 × 3 × 2 × 1)] = 4 × 126 = 504 ways. [1.5 Marks]

• **[Step 2: (b) At Least 3 Girls]** Case 1: 3 girls, 4 boys = 504 ways. Case 2: 4 girls, 3 boys = 4C4 × 9C3 = 1 x [(9 × 8 × 7) over (3 × 2 × 1)] = 1 × 84 = 84 ways. Total = 504 + 84 = 588 ways. [1.5 Marks]

• **[Step 3: (c) At Most 3 Girls]** Total committees of 7 from 13 people = 13C7 = 1716. At most 3 girls = Total - (4 girls) = 1716 - 84 = 1632 ways. [1 Mark]

• **[Boxed Result: (a) 504 ways, (b) 588 ways, (c) 1632 ways]**

7. Find the number of arrangements of the letters of the word 'PERMUTATIONS'. In how many of them: (a) Do words start with P and end with S? (b) Are vowels all together? (c) Are there always 4 letters between P and S? [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Letter Analysis & Part (a)]** 12 letters: T (2 times), others distinct. (a) Fix P first and S last. Remaining 10 letters have T twice: 10! over 2! = 1,814,400 ways. [1.5 Marks]

• **[Step 2: (b) Vowels All Together]** Vowels: E, U, A, I, O (5 vowels, all distinct). 7 consonants (with T twice) + 1 vowel block = 8 units. Arranged in 8! over 2! = 20,160. Vowels arranged inside block in 5! = 120 ways. Total = 20,160 × 120 = 2,419,200 ways. [1.5 Marks]

• **[Step 3: (c) 4 Letters Between P and S]** P and S can occupy positions: (1, 6), (2, 7), (3, 8), (4, 9), (5, 10), (6, 11), (7, 12) (7 positions). P and S can interchange (2 ways). Total ways for P and S = 7 × 2 = 14 ways. Remaining 10 letters arranged in 10! over 2! = 1,814,400 ways. Total = 14 × 1,814,400 = 25,401,600 ways. [1 Mark]

• **[Boxed Result: (a) 1,814,400; (b) 2,419,200; (c) 25,401,600]**